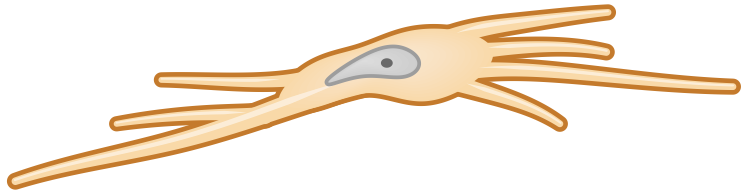
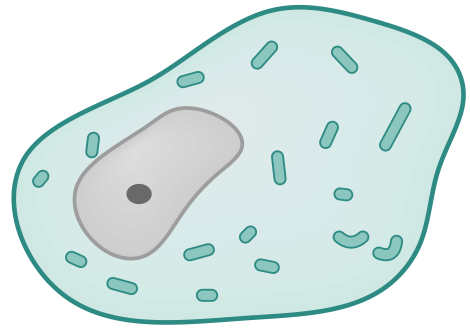
On the precision of dynamic gene regulation

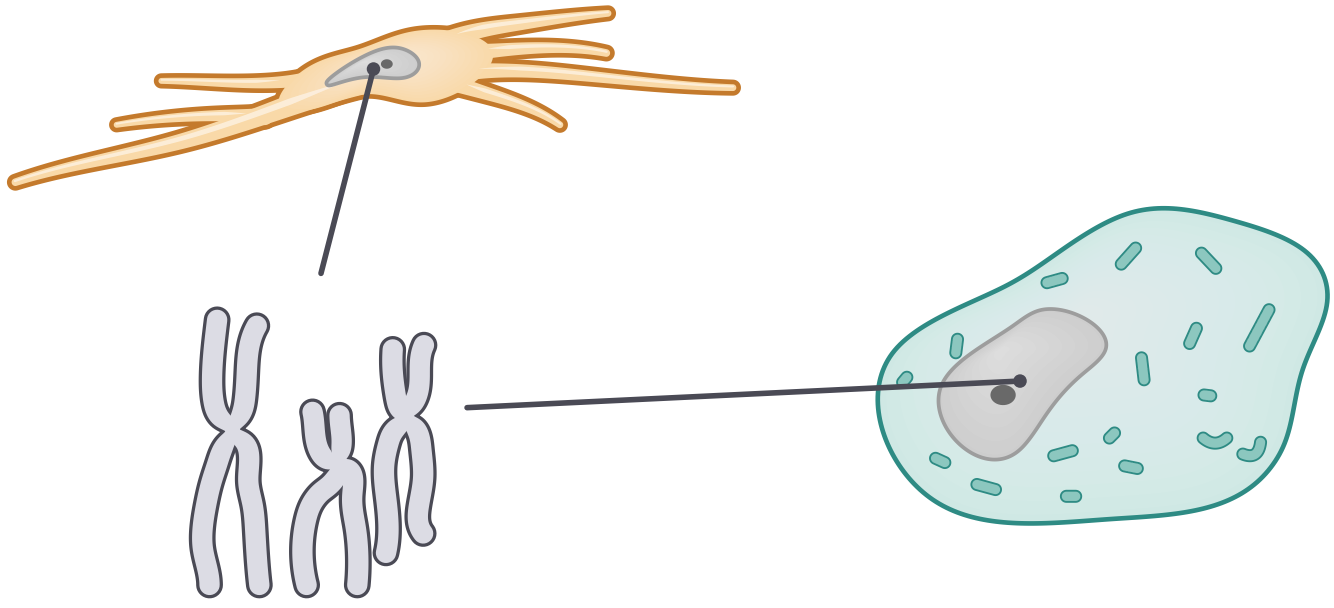
Tarek Tohme

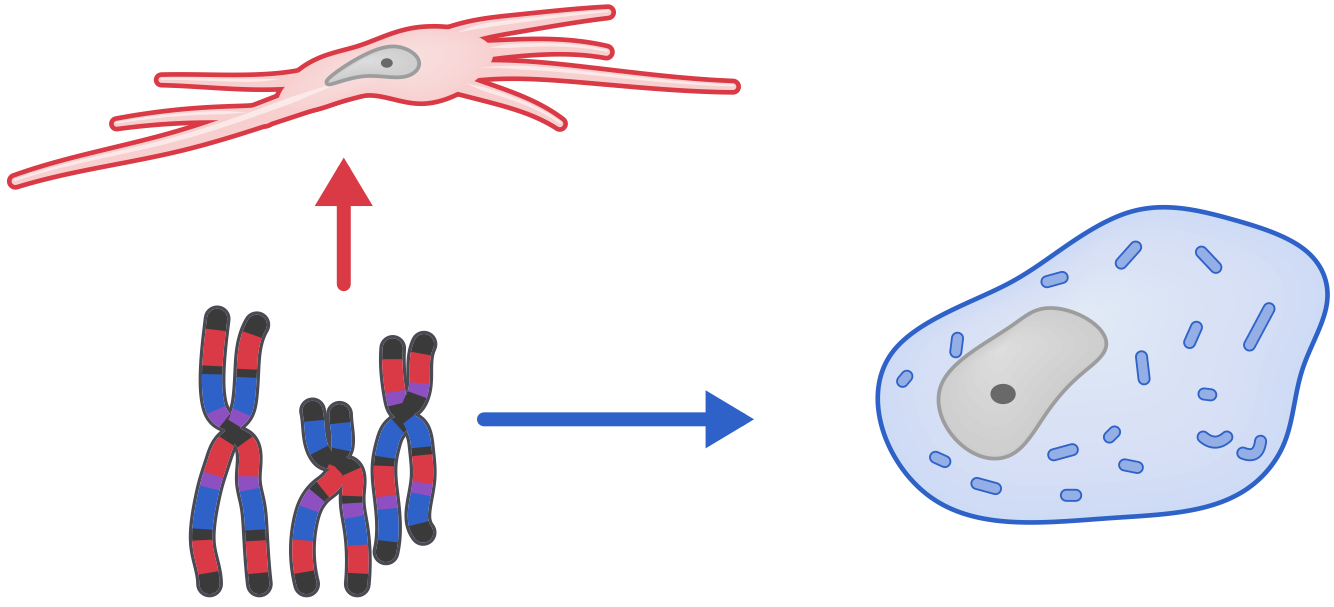
November 3, 2026

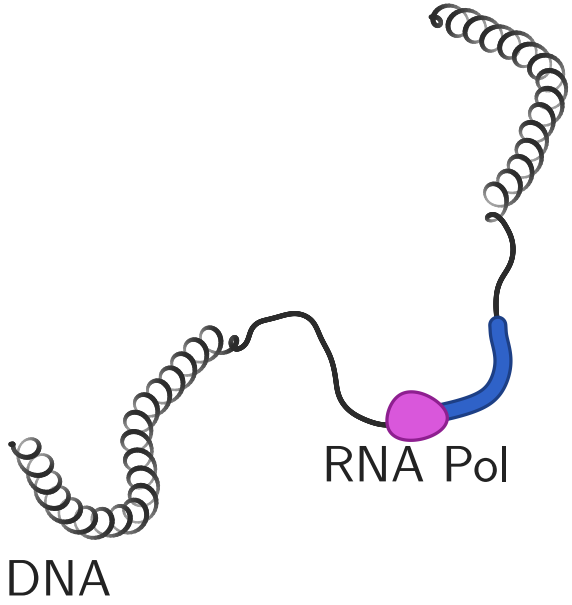




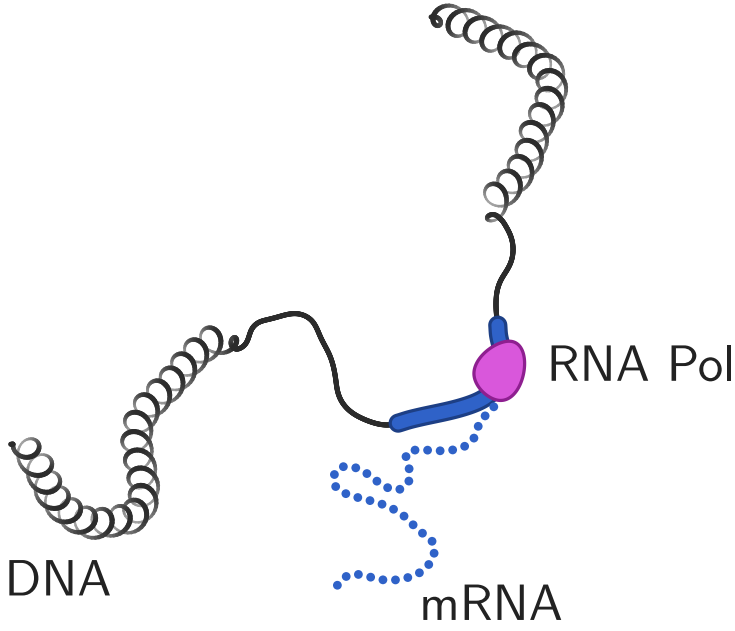




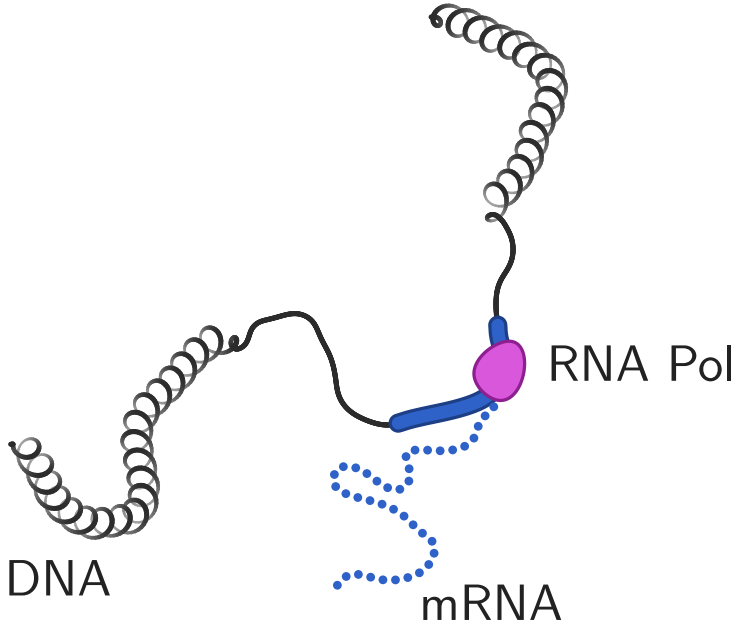




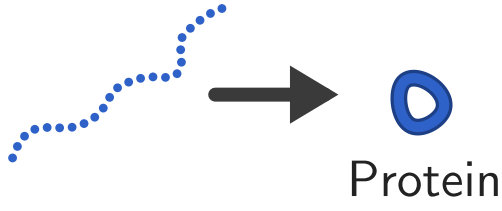
Transcription



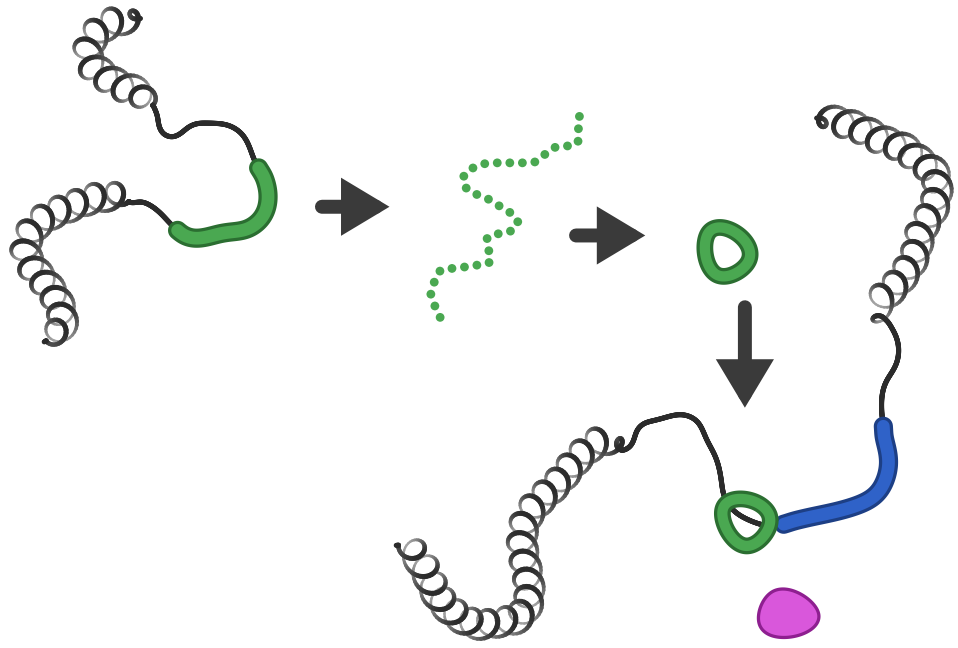
Transcription

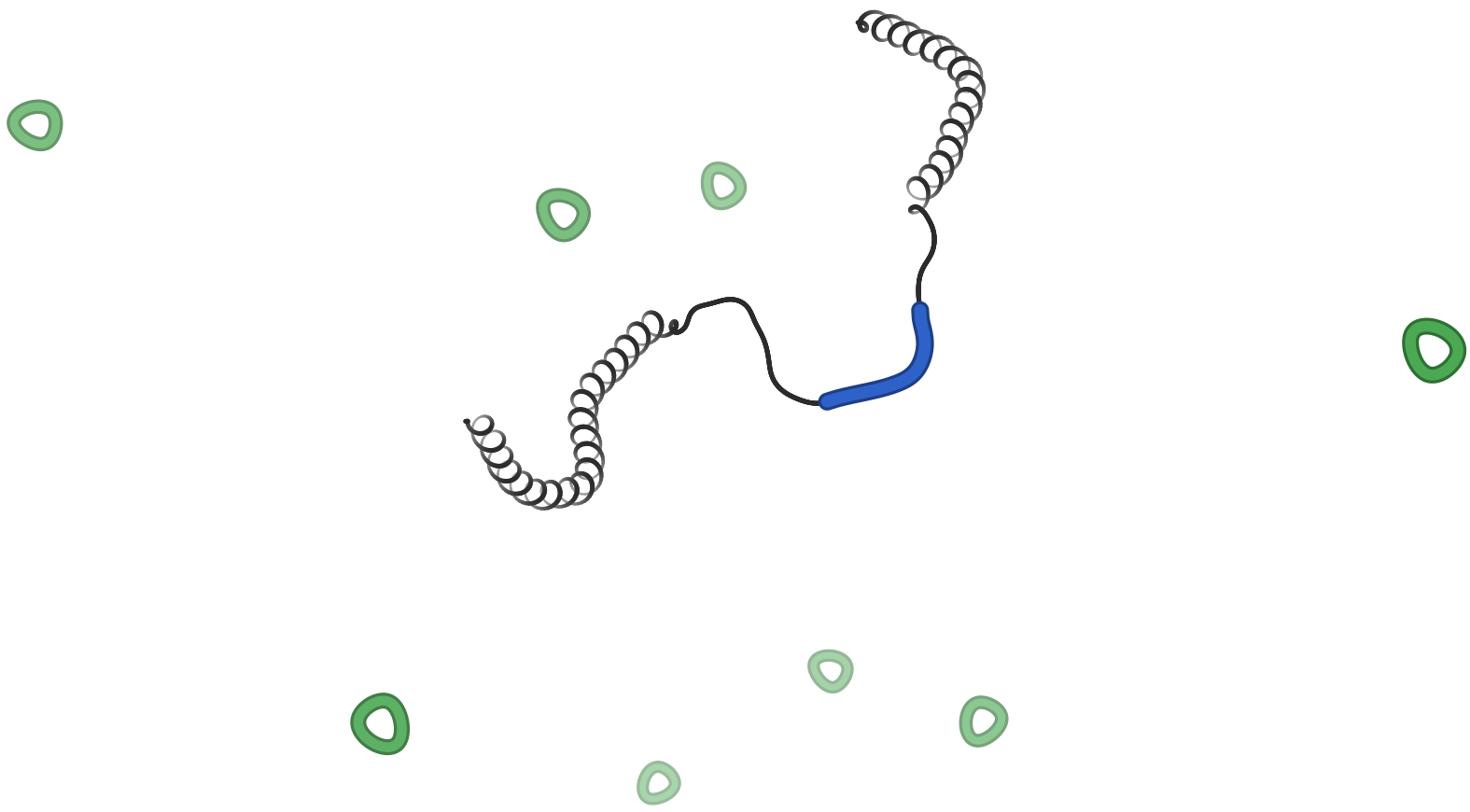


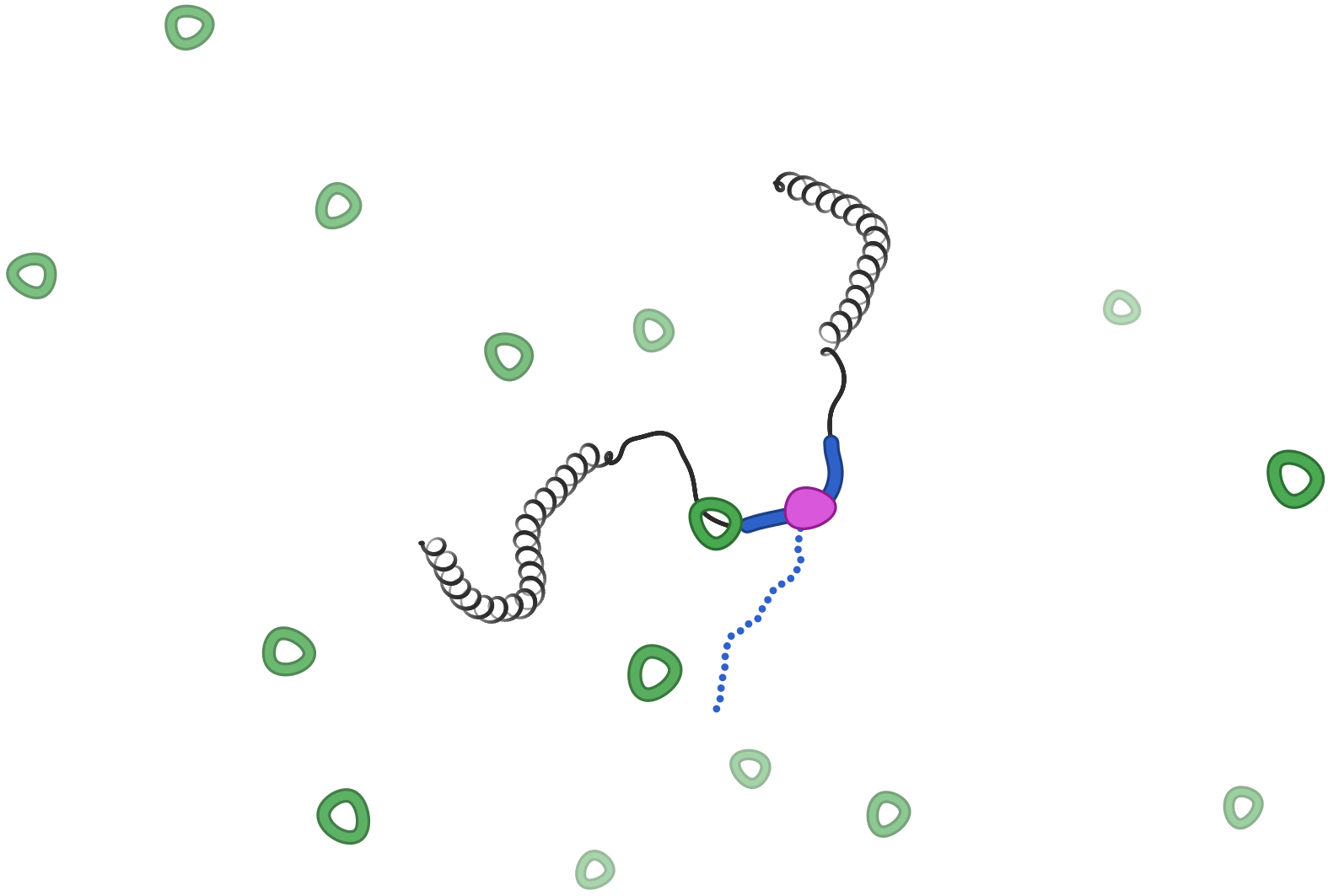
Transcription

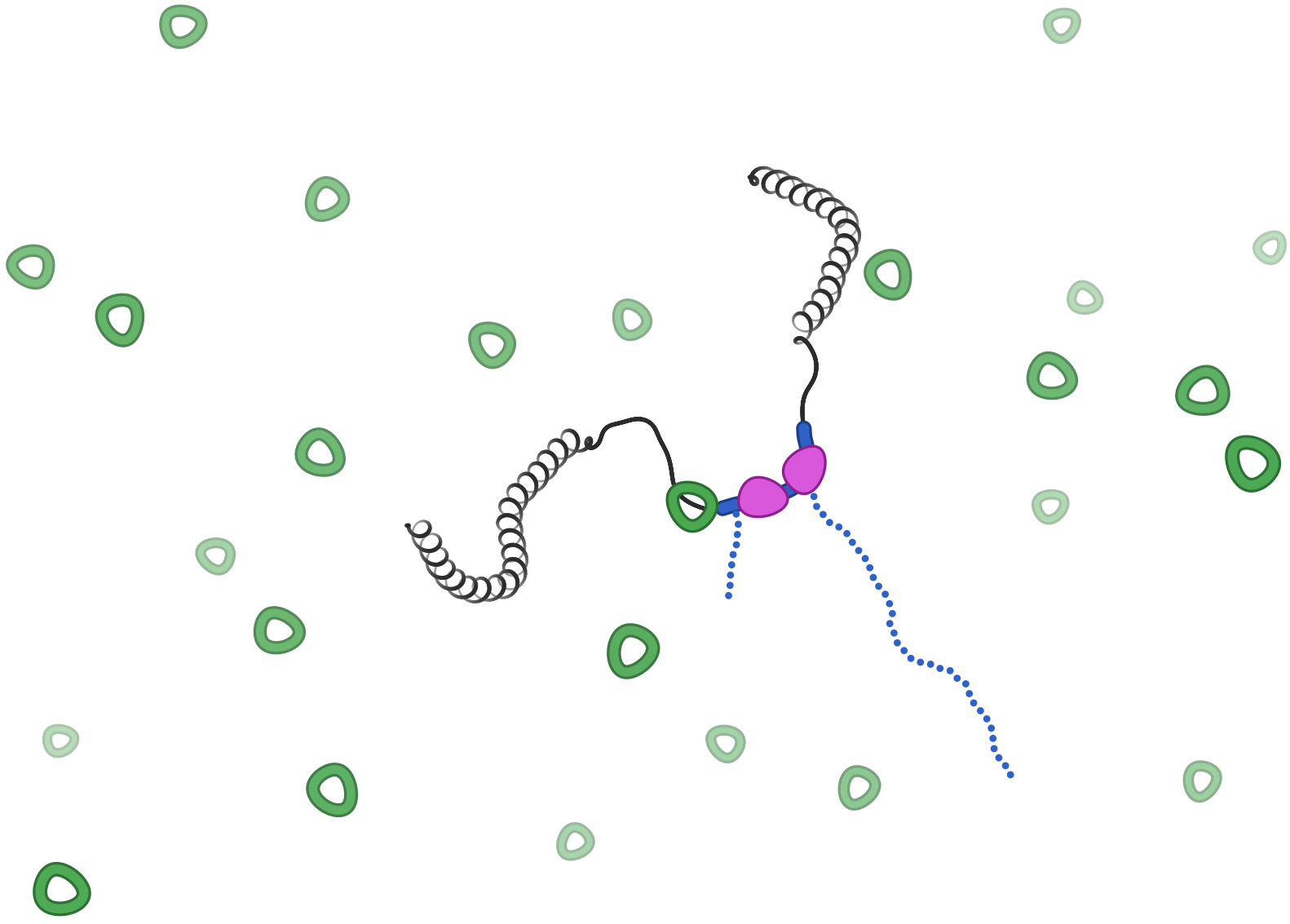


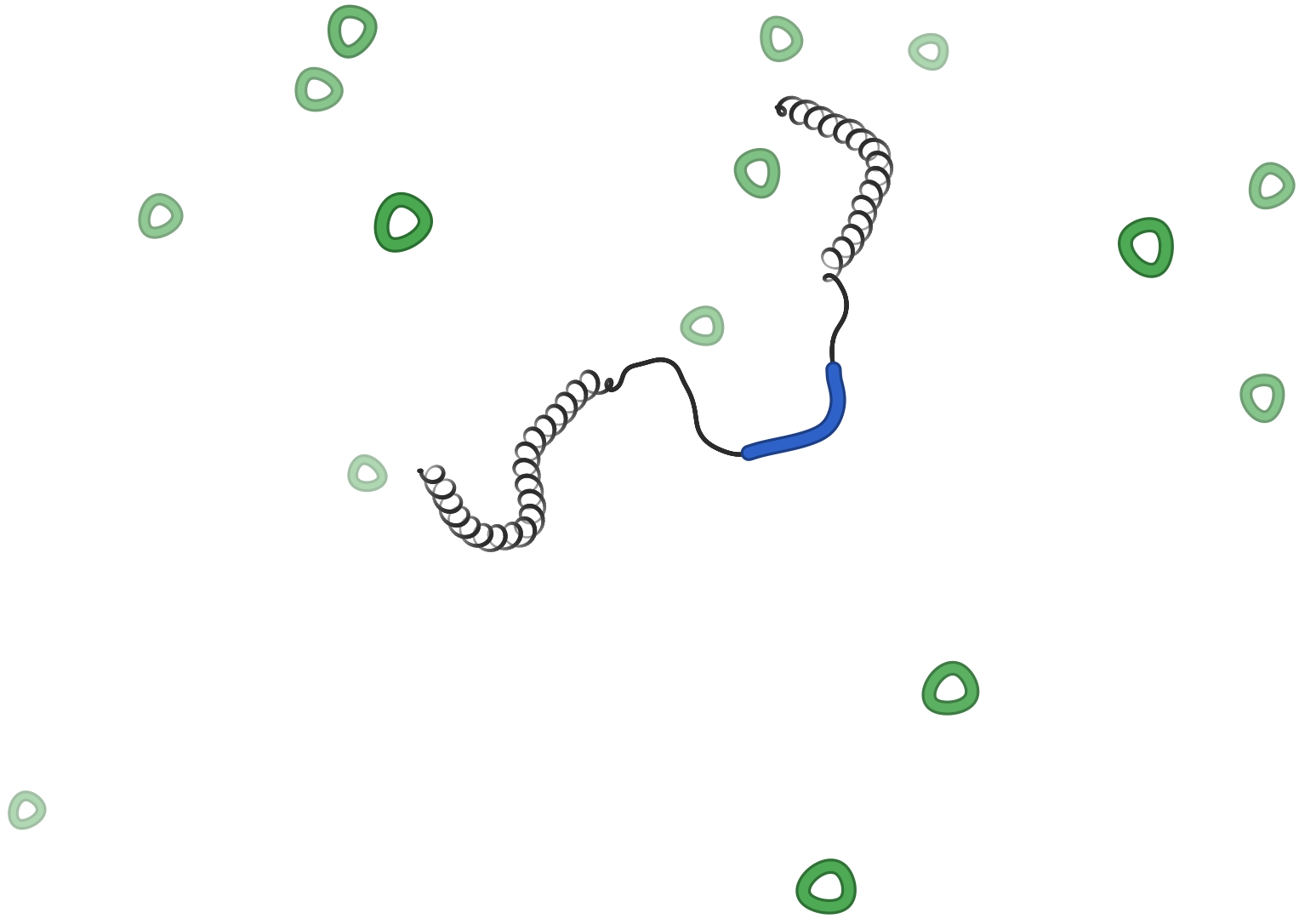
Translation

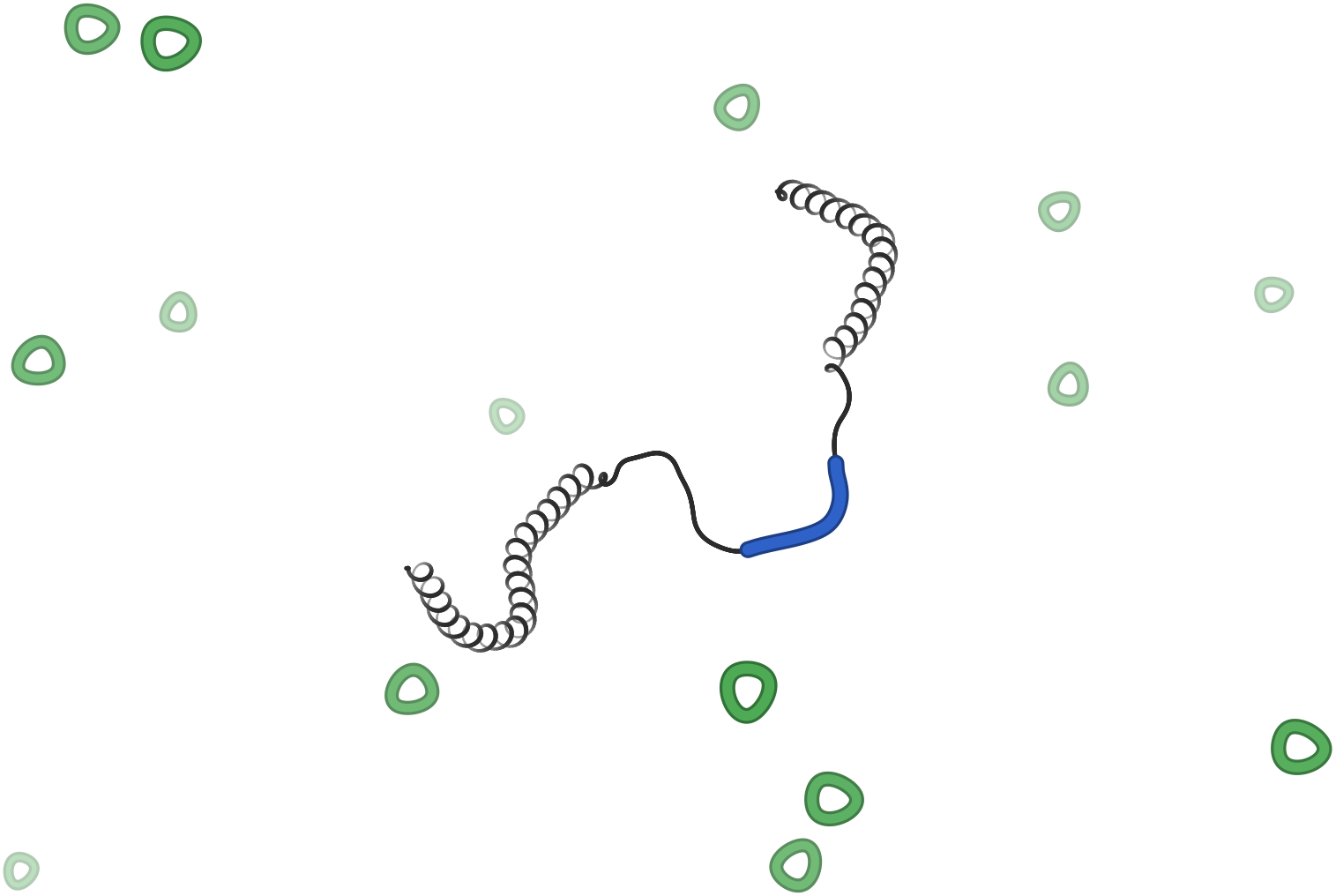


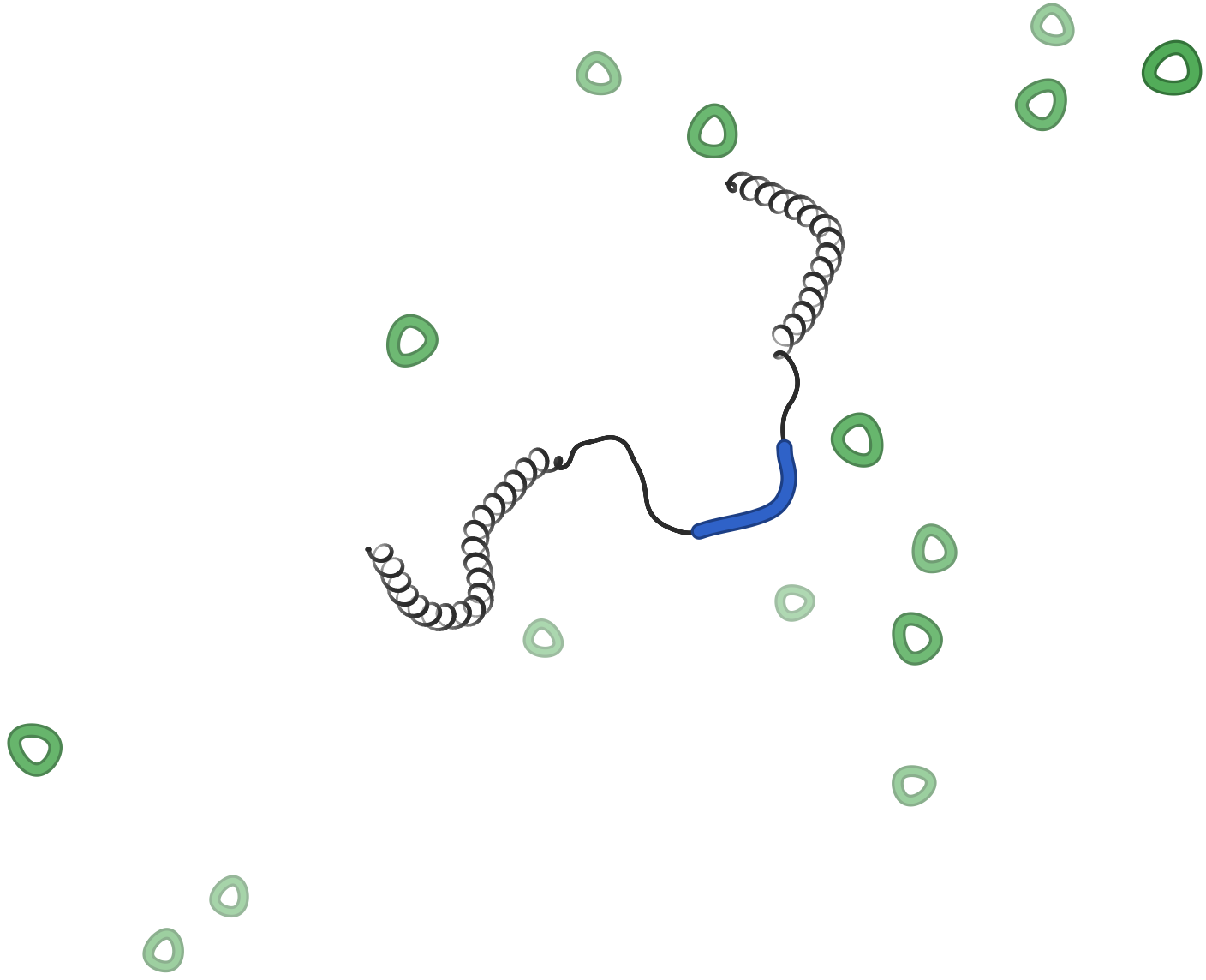


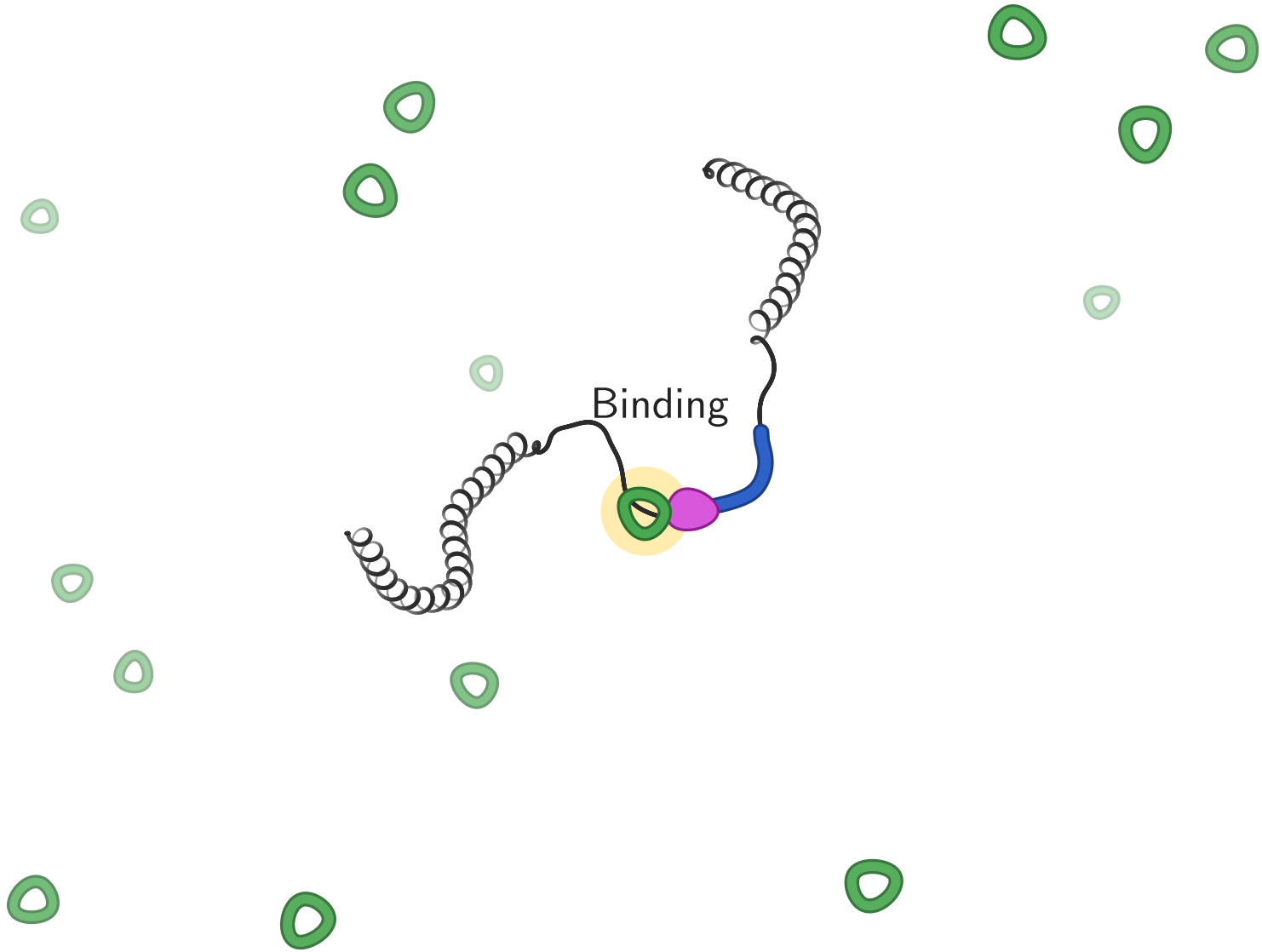


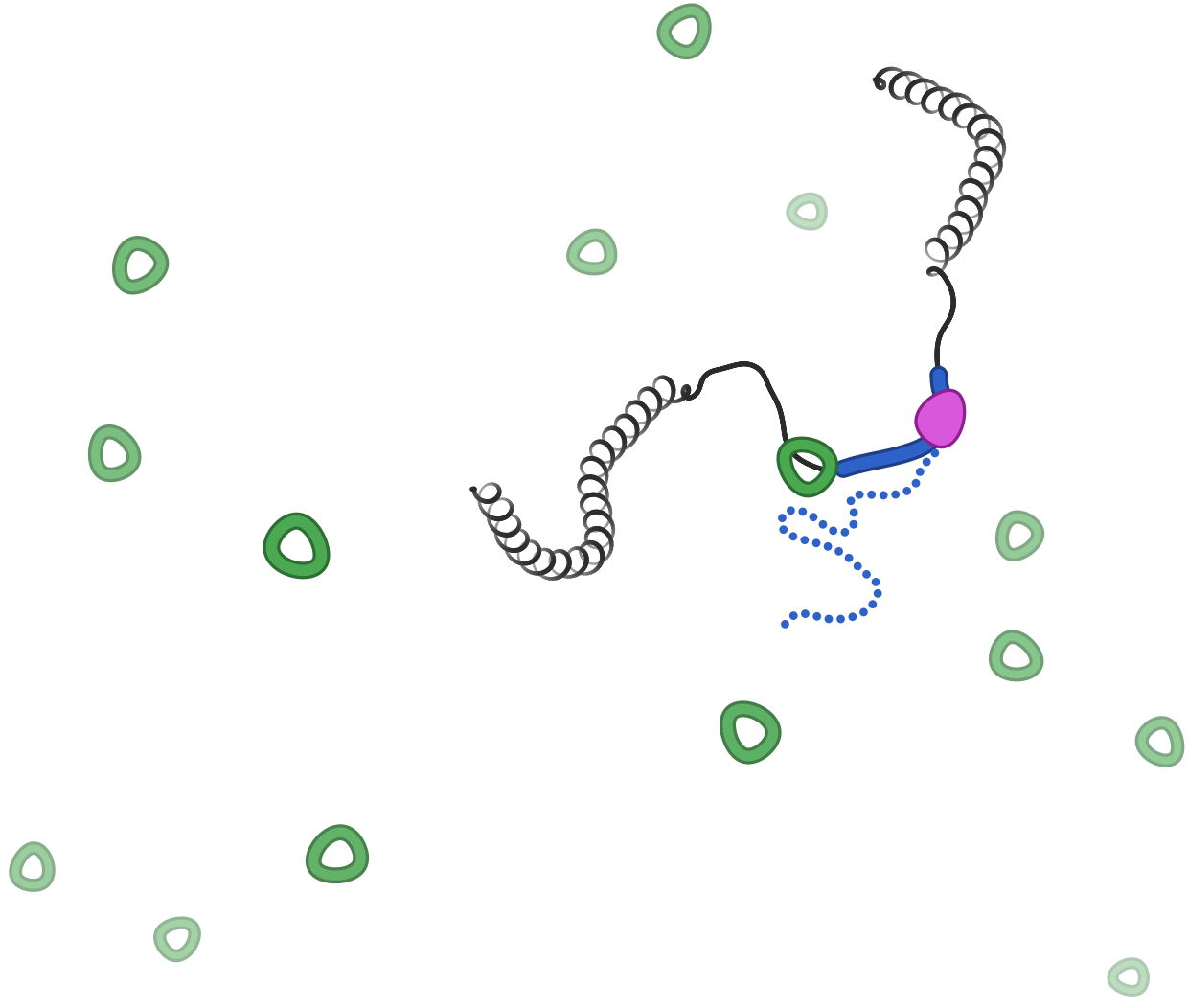


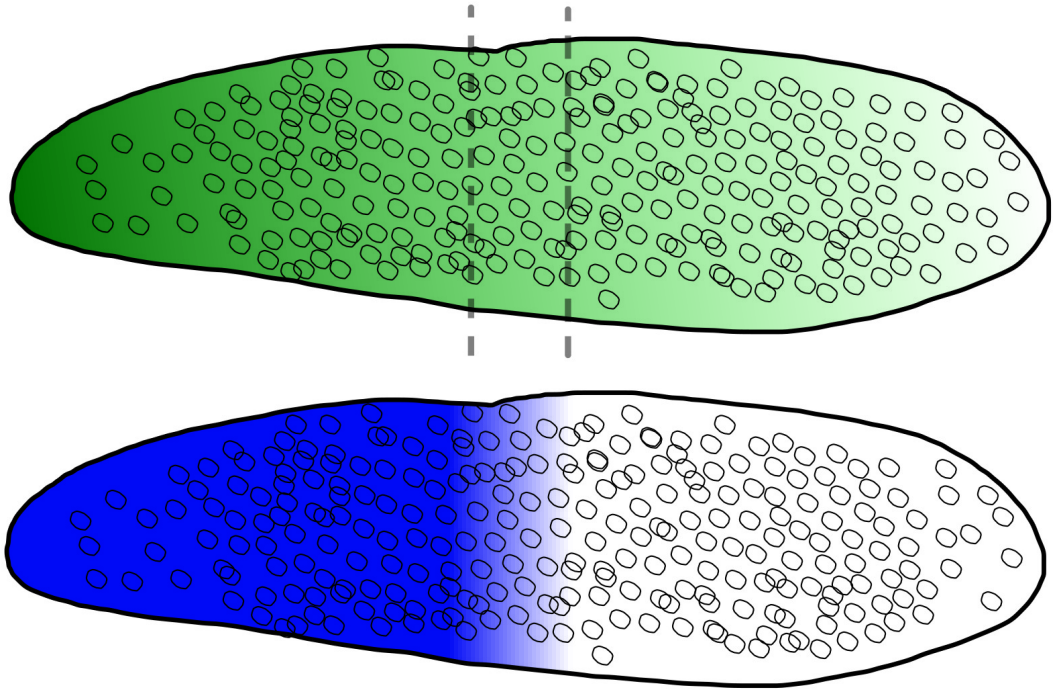






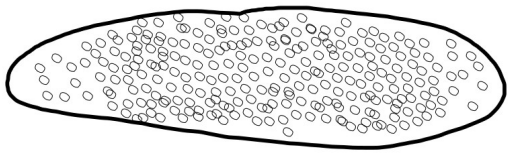
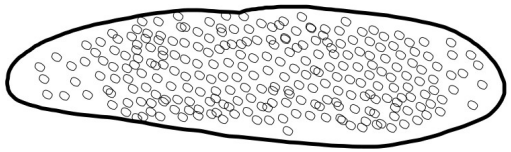

Binding



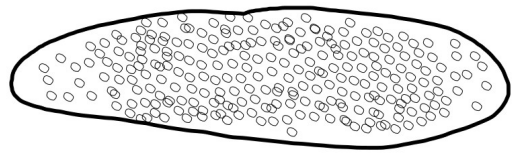
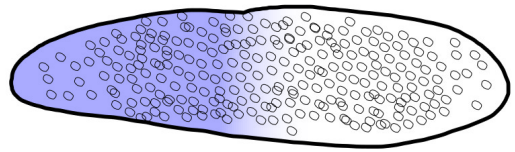


TF concentration

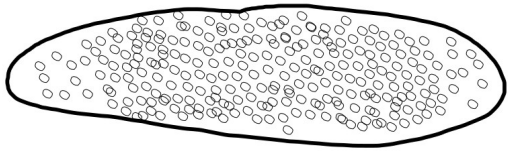
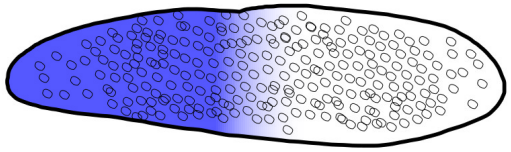
Level of expression
of controlled gene

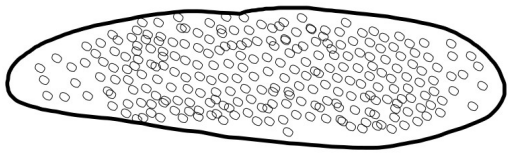
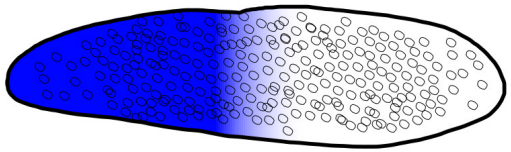
Division cycle	Embryo
1	
2	
3	
...	**...**
13	
14	

cycle duration:
4 minutes

Division cycle	Embryo
1	
2	
3	
...	**...**
13	
14	

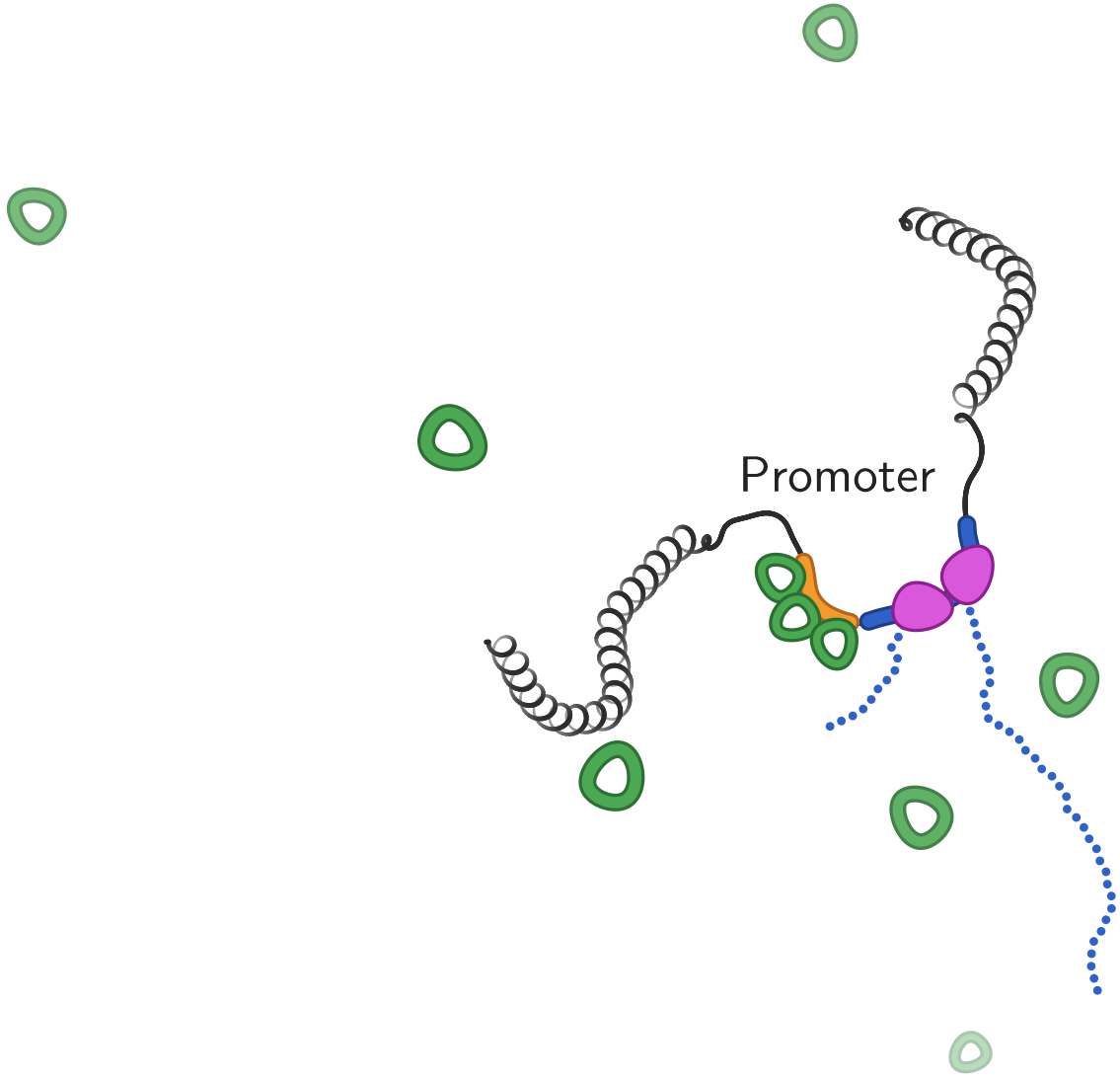
cycle duration:
4 minutes

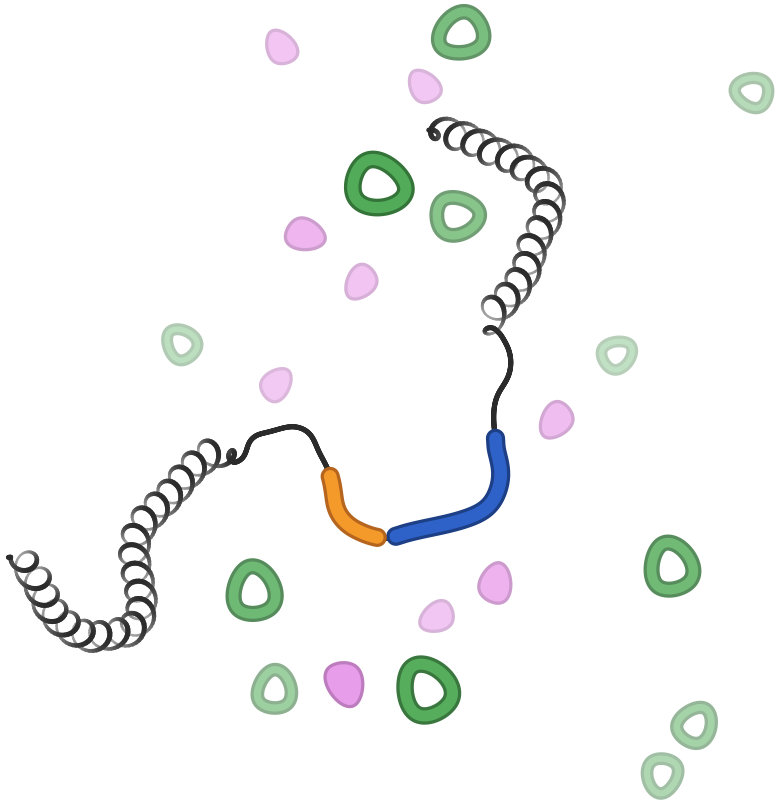
Division cycle	Embryo	
1		
2		
3		
...	**...**	
13		
14		cycle duration: **4 minutes**

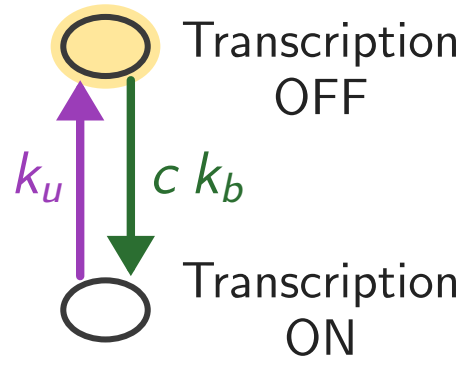
Division cycle	Embryo
1	
2	
3	
...	**...**
13	
14	

cycle duration:
4 minutes

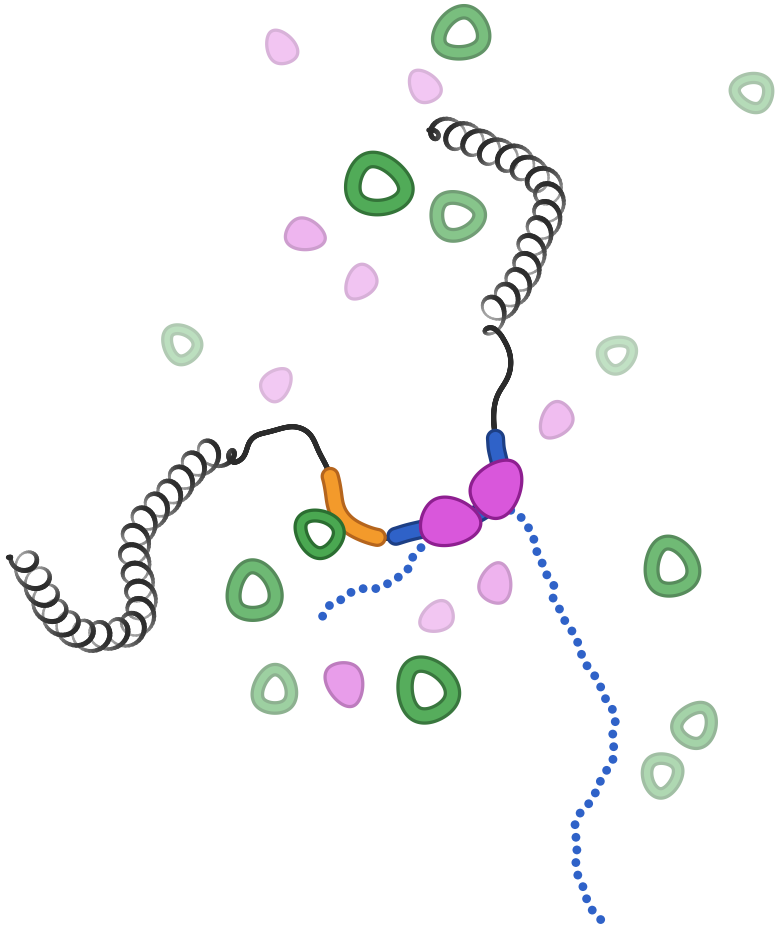


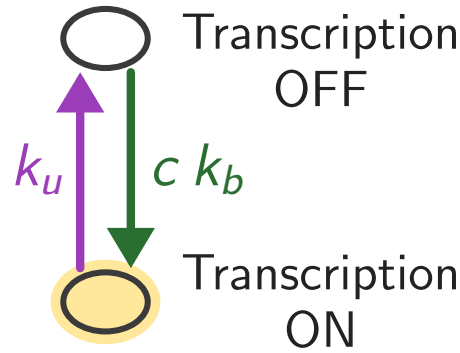

Promoter

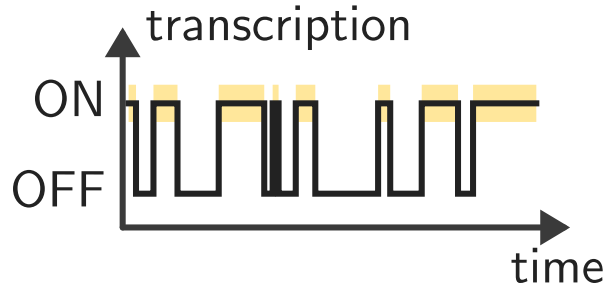


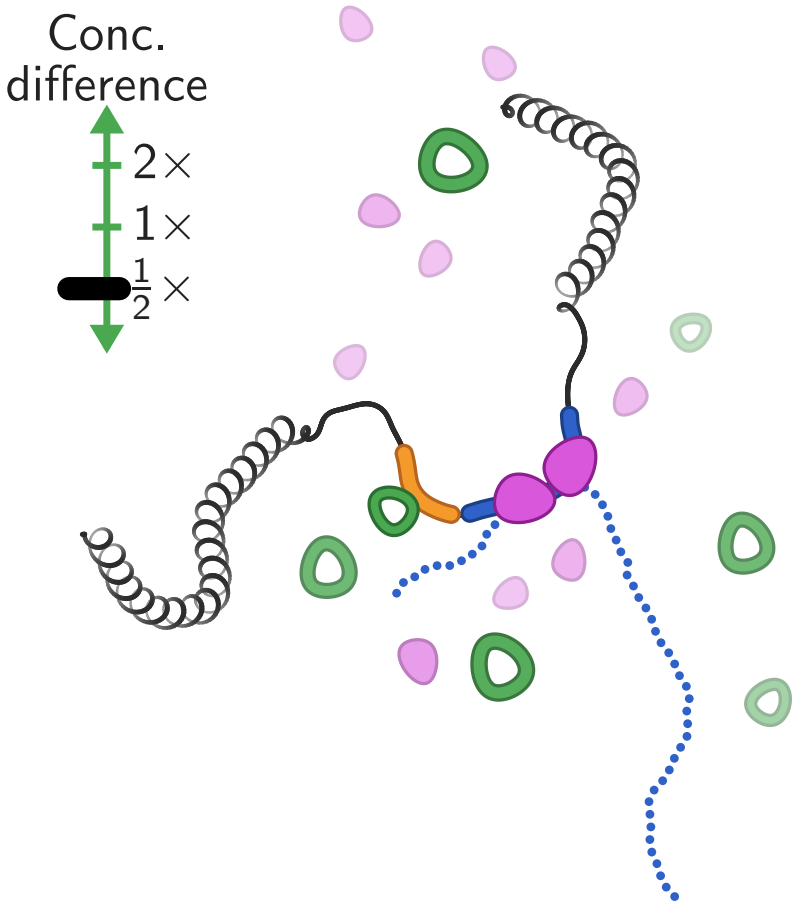

Transcription
OFF
k_u
$c\ k_b$
Transcription
ON

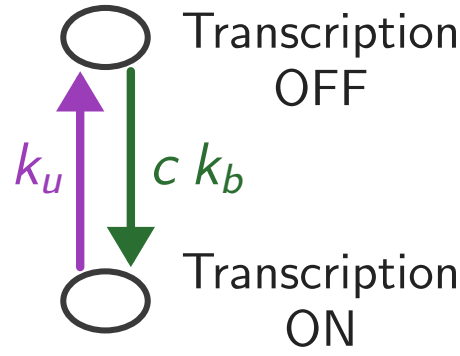

transcription
ON
OFF
time

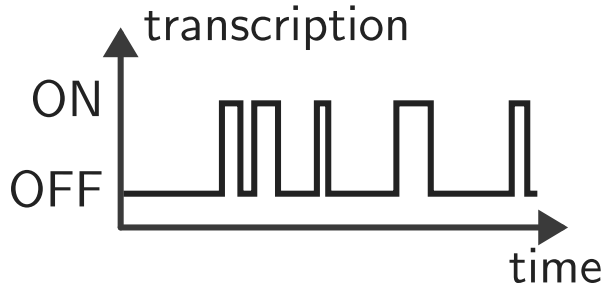


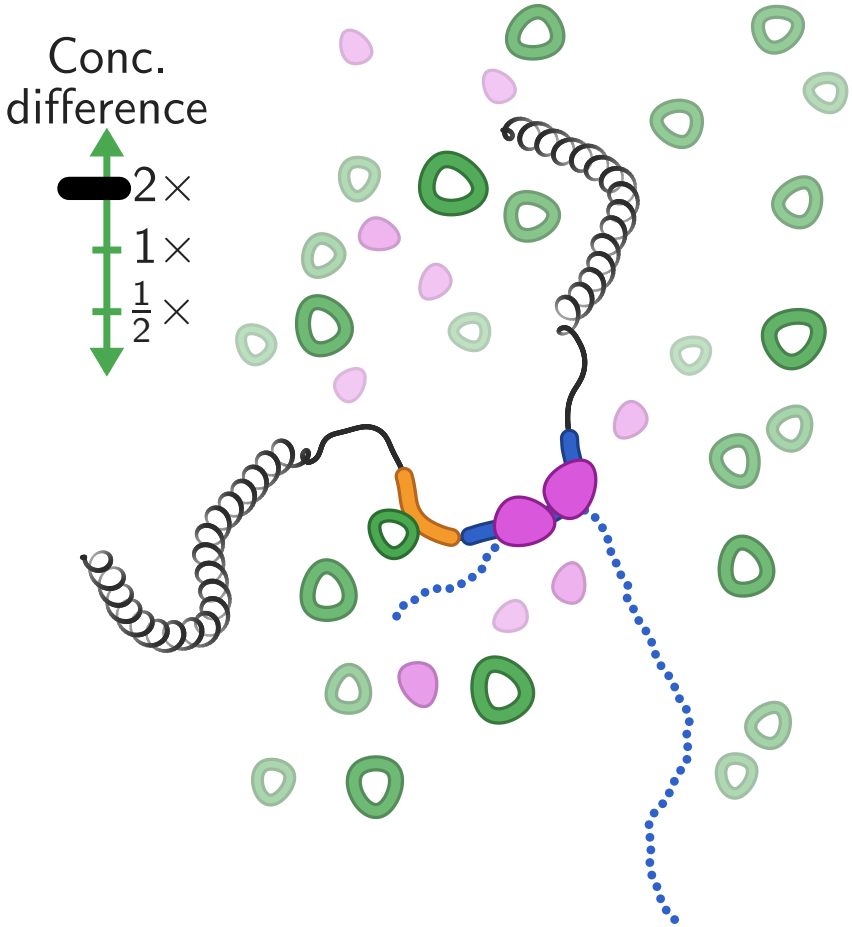

Transcription
OFF
k_u
$c\ k_b$
Transcription
ON

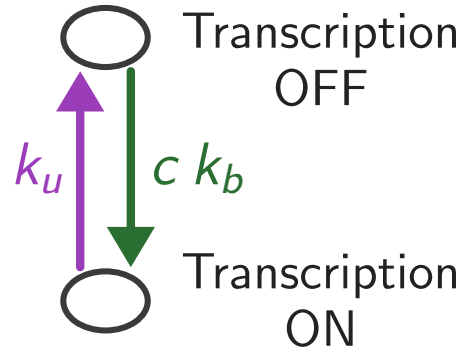

transcription
ON
OFF
time

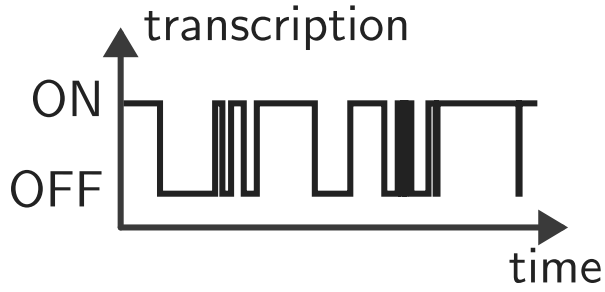

Conc.
difference
2×
1×
1/2 ×

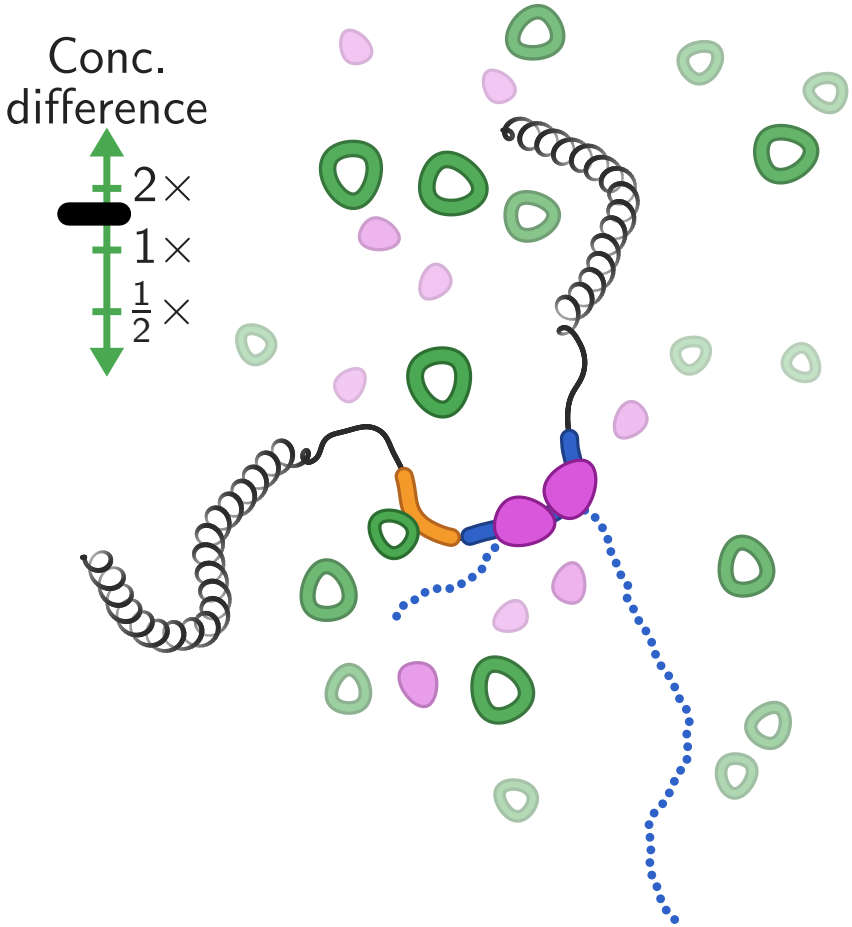

Transcription
OFF
k_u
$c\ k_b$
Transcription
ON

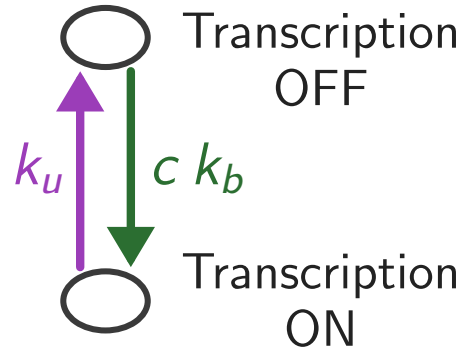

transcription
ON
OFF
time

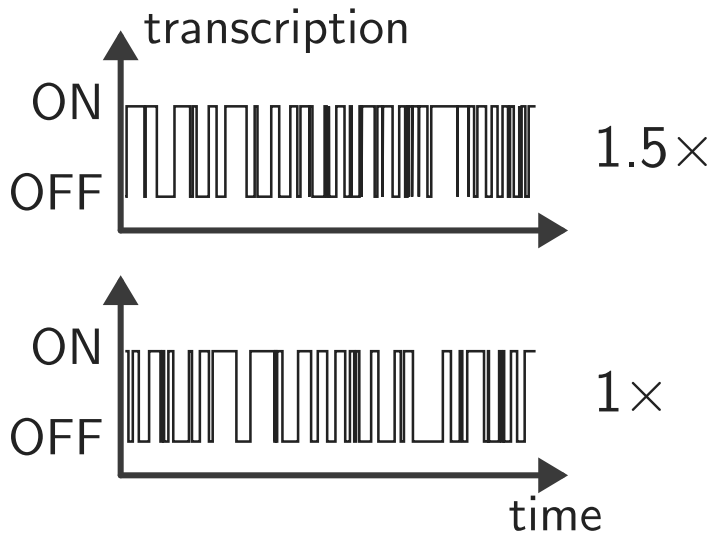

Conc.
difference
2×
1×
1/2×

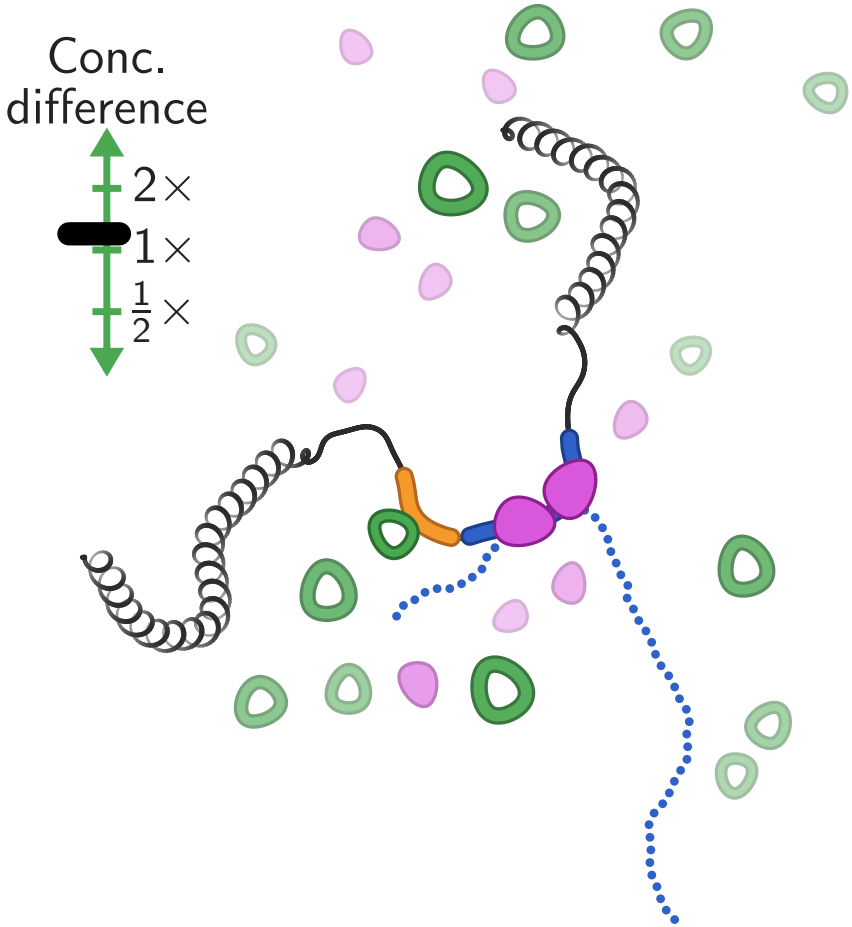

Transcription
OFF
k_u
$c\,k_b$
Transcription
ON

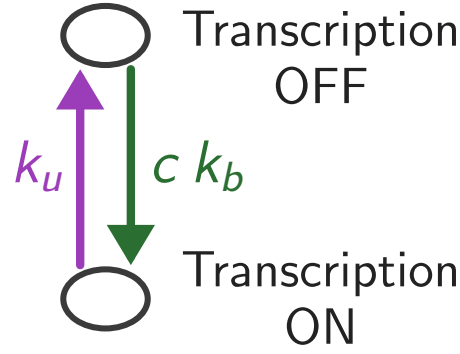

transcription
ON
OFF
time

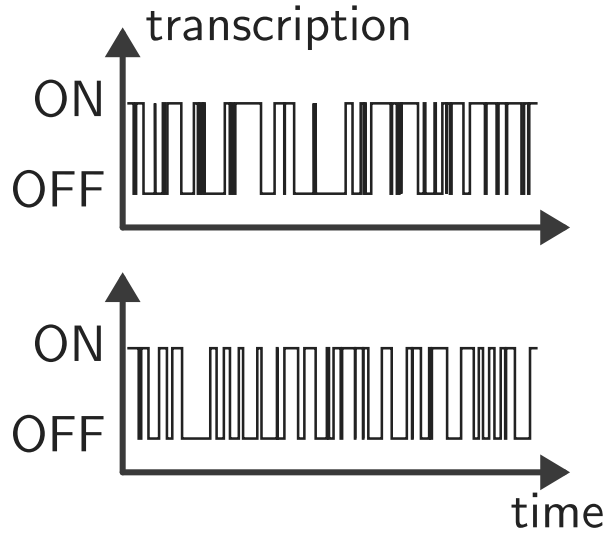

Conc.
difference
2×
1×
1/2×


Transcription
OFF
k_u
$c\,k_b$
Transcription
ON


transcription
ON
OFF
1.5×
ON
OFF
1×
time


Conc.
difference
2×
1×
1/2×


Transcription
OFF
k_u
$c\ k_b$
Transcription
ON


transcription
ON
OFF
ON
OFF
time

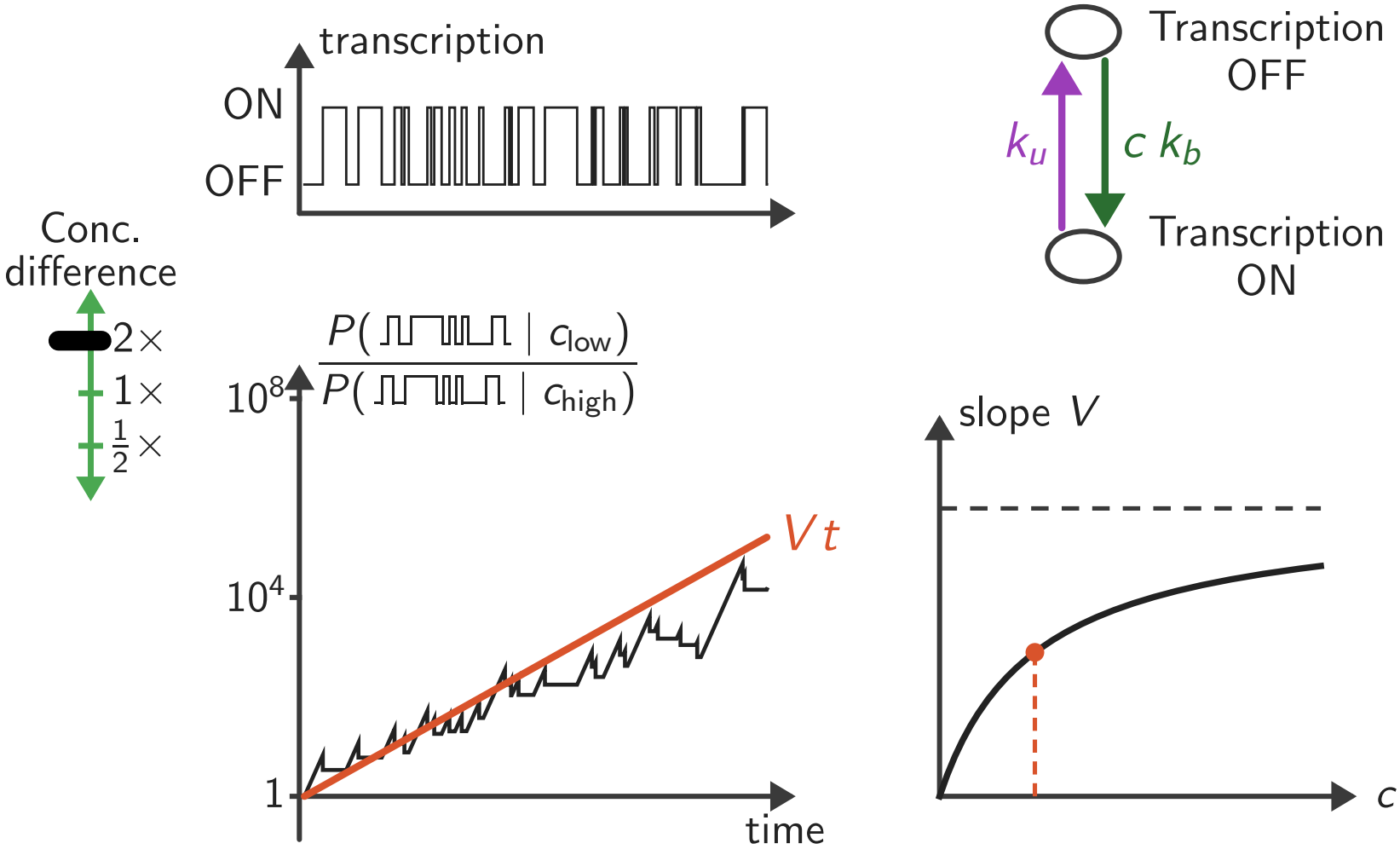
$$P(\text{  } \mid c_{\text{low}}) \overset{?}{>} P(\text{} \mid c_{\text{high}})$$

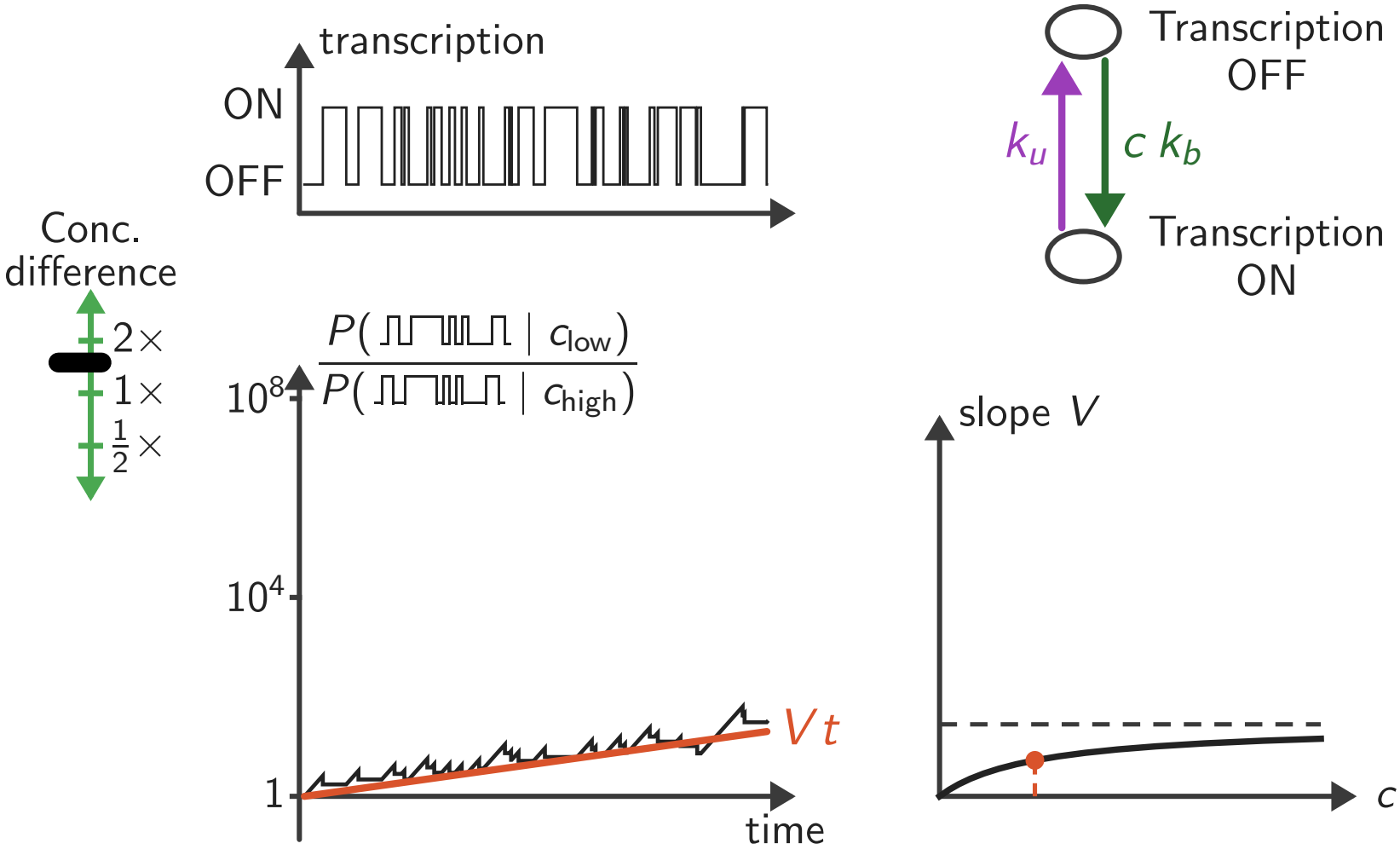
$$\frac{P(\text{} \mid c_{\text{low}})}{P(\text{} \mid c_{\text{high}})} > 1$$

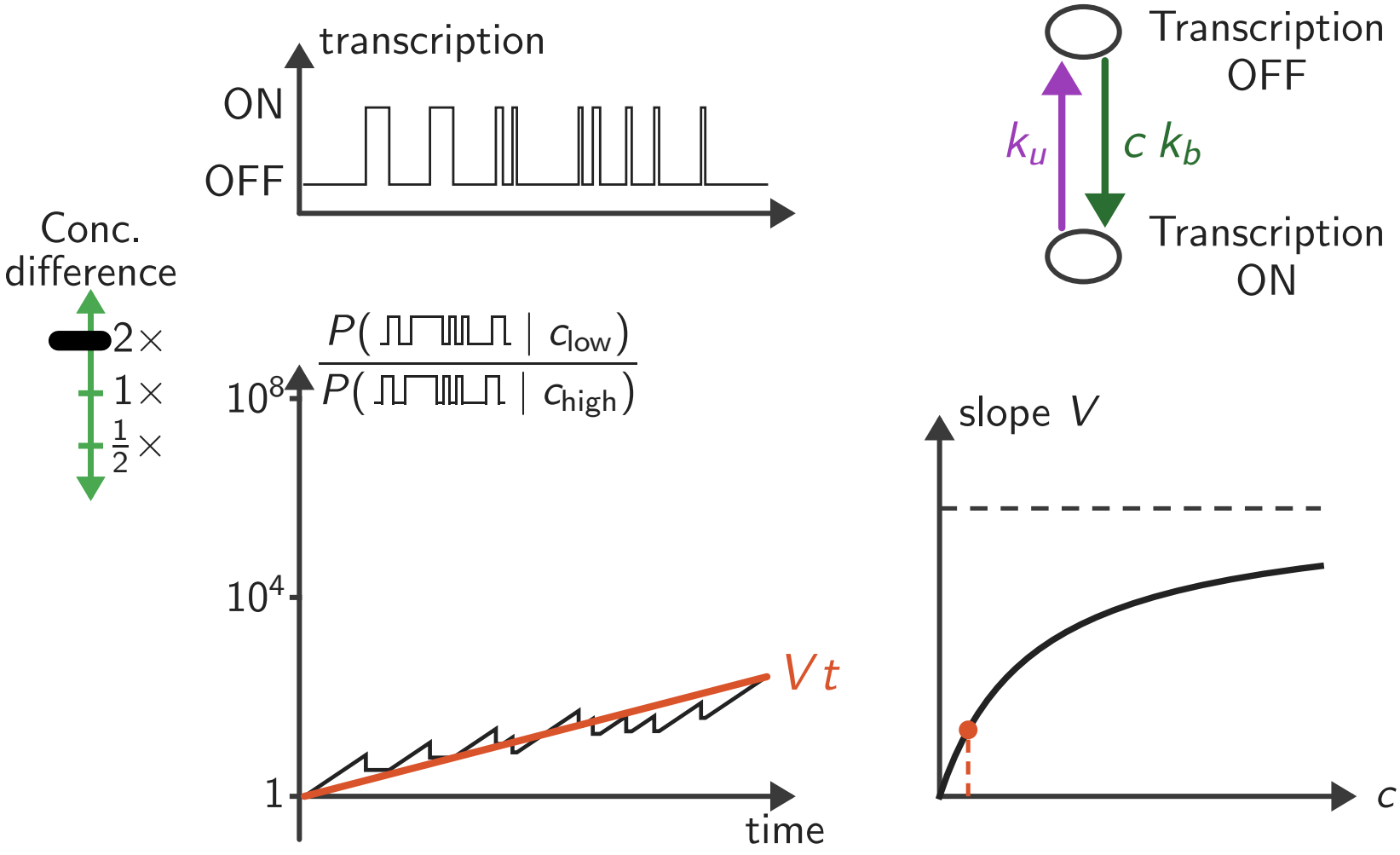
$$P(\ \text{}\ \mid c_{\text{low}}) \overset{?}{>} P(\ \text{}\ \mid c_{\text{high}})$$

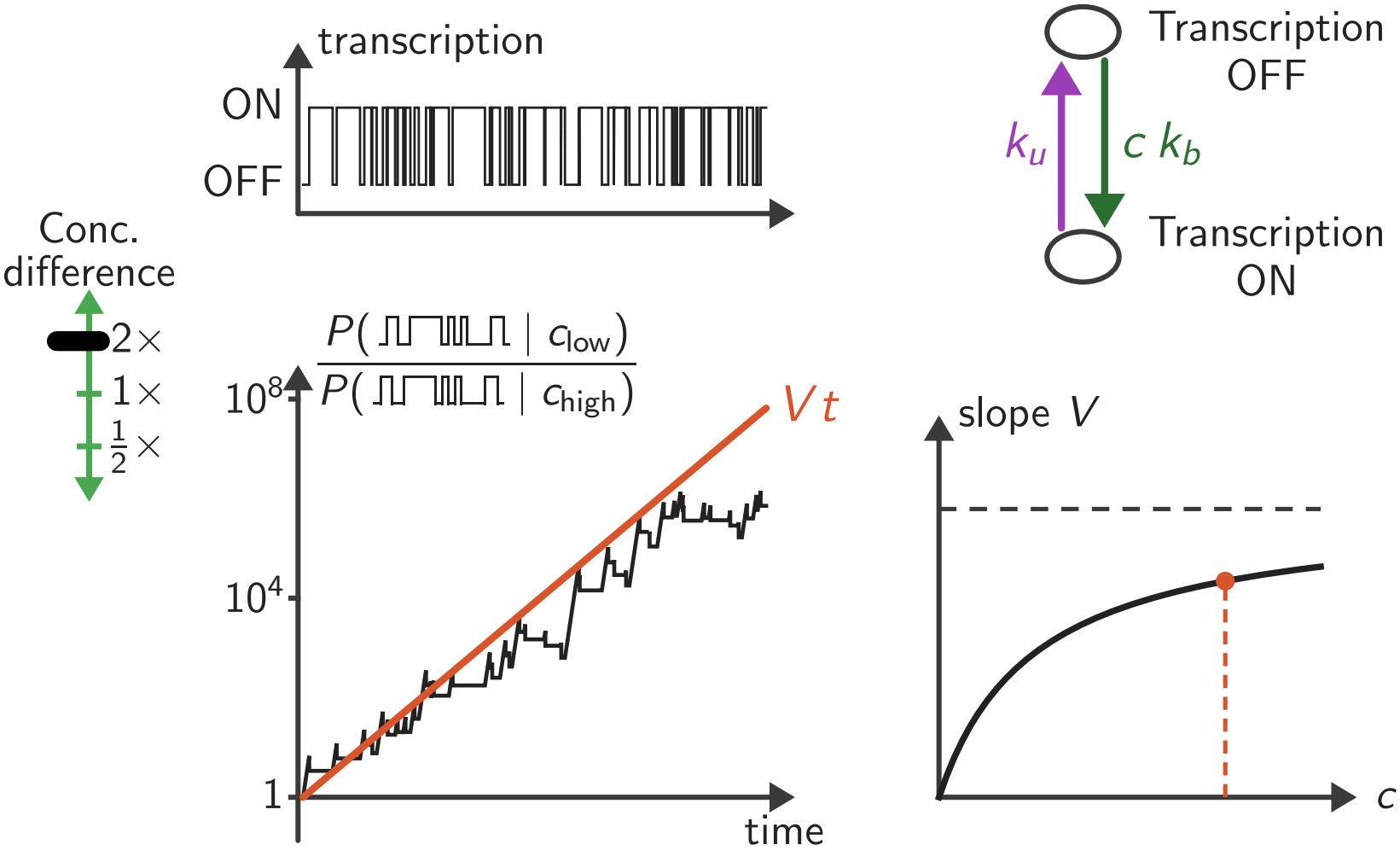
$$R(t) = \frac{P(\ \text{}\ \mid c_{\text{low}})}{P(\ \text{}\ \mid c_{\text{high}})} > 1$$

$$\log R(t) = V t + \sqrt{2D}\,\xi$$


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓} \mid c_{\text{high}})}$
10^8
10^4
1
Vt
time
Transcription OFF
k_u
$c\,k_b$
Transcription ON
slope V
c


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓⊓} \mid c_{\text{high}})}$
10^8
10^4
1
Vt
time
Transcription OFF
k_u
$c\,k_b$
Transcription ON
slope V
c


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓} \mid c_{\text{high}})}$
10^8
10^4
1
Vt
time
Transcription OFF
k_u
$c\,k_b$
Transcription ON
slope V
c


transcription
ON
OFF
Conc. difference
2×
1×
1/2×
$\frac{P(\text{⊓⊓⊓} \mid c_{\text{low}})}{P(\text{⊓⊓⊓} \mid c_{\text{high}})}$
10^8
10^4
1
time
Vt
Transcription OFF
k_u
$c\,k_b$
Transcription ON
slope V
c